

Phase 4: Project Planning

4.1 Project Milestones

Milestone	Activities	Deliverable
M1: Model Selection & Architecture	Research Gemini models, define architecture, set up environment and <code>.env</code>	Architecture diagram, working environment
M2: Core Functionalities	Generation logic, formatting modules (TXT/DOCX/PDF/HTML), editable preview design	Core modules in <code>ai_core/</code>
M3: API Logic Integration	<code>main.py</code> , <code>routes.py</code> , <code>DocumentRequest</code> , <code>/generate</code> endpoint	Working backend API
M4: Frontend Development	Streamlit UI, preview, edit, downloads, branding	Working web UI
M5: Deployment (local)	<code>requirements.txt</code> , run backend + frontend, verify end-to-end	Running application
M6: Testing, Documentation, Demo	Test cases, docs, GitHub upload, demo video	Final submission

4.2 Work Breakdown Structure

ID	Task	Depends on	Estimate
T1	Study FastAPI, Streamlit and Gemini API docs	none	1 day
T2	Set up Python env, virtual env, install packages	T1	0.5 day
T3	Get Gemini API key, create <code>.env</code> , <code>config.py</code>	T2	0.5 day
T4	Implement <code>GeminiDocumentGenerator</code> and prompt	T3	1 day
T5	Implement <code>main.py</code> and <code>routes.py</code>	T4	1 day
T6	Implement <code>sanitize_text</code> , <code>format_html_preview</code>	T4	0.5 day
T7	Implement <code>format_docx</code>	T6	1 day
T8	Implement <code>format_pdf</code>	T6	1 day
T9	Build Streamlit UI (inputs, preview, edit)	T5, T6	1.5 days
T10	Add download buttons and branding	T7, T8, T9	1 day
T11	Testing and bug fixing	T10	2 days
T12	Write phase-wise documentation, screenshots	T11	1.5 days
T13	Record demo video, final GitHub upload	T12	1 day

4.3 Timeline (Sample 4-Week Plan)

Week	Focus	Output
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Week 1	Ideation, requirements, design, environment setup (T1-T3)	Phases 1-3 docs, environment ready
Week 2	Backend and AI integration (T4-T6)	/generate works via Swagger UI
Week 3	Formatting modules and frontend (T7-T10)	End-to-end app working locally
Week 4	Testing, documentation, demo (T11-T13)	Final GitHub repo + demo video

Adjust the dates to your team's schedule.

4.4 Team Roles and Responsibilities

Member	Role	Responsibilities
	Team Lead / Backend	FastAPI, routes, integration, GitHub management
	Frontend	Streamlit UI, preview, edit, downloads
	AI / Formatting	Prompt design, Gemini integration, DOCX/PDF modules
	QA / Documentation	Test cases, screenshots, phase docs, demo video

(Team size: minimum 2, standard 4, maximum 5. Merge roles for smaller teams.)

4.5 Tools and Collaboration

Purpose	Tool
Code	VS Code, Python 3.10+

Version control	Git and a public GitHub repository
API testing	FastAPI Swagger UI (/docs), Postman
Communication	WhatsApp / Google Meet / Discord
Demo recording	Screen recorder with voice-over, upload to Google Drive (public link)

4.6 Git Workflow

- `main` branch holds stable code; each member works on a feature branch (`feature/backend`, `feature/frontend`, `feature/formatting`)
- Small, meaningful commits; merge through pull requests or reviewed merges
- Never commit `.env`, `venv/`, or `__pycache__/`

4.7 Risk Management

Risk	Likelihood	Impact	Mitigation
Gemini API quota / rate limit errors	Medium	High	Retry with backoff, friendly error messages, keep test inputs short
Model name deprecated / unavailable	Medium	High	Keep model name in config; confirm in Google docs before demo
API key leaked to GitHub	Low	High	<code>.env</code> in <code>.gitignore</code> ; rotate key if exposed
PDF fails on special characters	Medium	Medium	<code>sanitize_text</code> ; Latin-1 safe encoding; optionally use <code>fpdf2</code> with Unicode font

Inconsistent AI output (markdown symbols, placeholders)	High	Medium	Prompt guardrails, cleanup in formatters, user edit step
Inaccurate legal content	Medium	High	Disclaimer in UI/exports; encourage lawyer review
Team member unavailable	Medium	Medium	Shared repo, documented code, cross-training

4.8 Definition of Done

- All functional requirements FR-1 to FR-12 implemented and tested
- All three export formats verified
- Code and 8 phase documents pushed to public GitHub repo
- Demo video recorded and shared publicly via Google Drive